Software Requirements Specification

for

FamPlan

Version 1.6 - Lab 5 Final Revision

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NTU SC2006 Software Engineering

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Revision History

Name	Date	Reason For Changes	Version
Banana Five-O	8 September 2026	Initial SRS for Lab 1 submission	1.0
Banana Five-O	12 September 2026	Revised SRS for family-neighbourhood scope	1.1
Banana Five-O	5 October 2026	Revised SRS for Lab 3 submission	1.2
Banana Five-O	16 October 2026	Design refinement: 9.4 design layer, 9.5 traceability	1.5
Banana Five-O	31 October 2026	Lab 5: travel mode per destination; commute scored across all destinations; larger architecture figure; class diagrams updated	1.6

1. Introduction

1.1 Purpose

This Software Requirements Specification defines FamPlan as a family-neighbourhood decision-support web application. It helps couples and households planning for children compare Singapore neighbourhoods using nearby childcare, kindergartens, primary and secondary schools, supermarkets, healthcare, green spaces, public transport, commute estimates, and accessibility. The product focuses exclusively on family-neighbourhood discovery and comparison. This revision records 48 functional requirements, 29 measurable non-functional requirements, a tabular data dictionary, a 16-use-case model, traceability, and a simple technology recommendation.

1.2 Document Conventions

Functional requirements use shall and have unique FR identifiers. Non-functional requirements use must or must not and have unique NFR identifiers. Must and Should are priority classifications. Visitor is the base public actor. Registered User inherits Visitor behaviour, and Data Administrator inherits Registered User behaviour. Automatic data synchronisation is internal system behaviour; administrators monitor its status and do not manually start updates.

1.3 Intended Audience and Reading Suggestions

The document is intended for the project team, teaching staff, developers, testers, and future maintainers. Sections 1 and 2 define the problem and scope, Section 3 records interfaces, Sections 4 and 5 are the testable requirement baseline, Section 6 defines shared domain terms, Section 7 shows use-case coverage, and Section 8 proposes a deliberately small implementation stack.

1.4 Product Scope

FamPlan helps a household answer a focused question: which neighbourhood best supports the family's planned life stages and daily needs? Users search neighbourhoods, inspect family-relevant facilities and travel information, adjust transparent weights, compare alternatives, and save a shortlist. The score is an explainable comparison aid rather than an instruction or guarantee. FamPlan is not a property marketplace, valuation or transaction service, school-admission predictor, live childcare-vacancy service, medical adviser, or guarantee that one area suits every family.

1.5 References

Reference	Purpose
SC2006 Lab 1 Manual	Defines requirements, data dictionary, initial use-case model, and UI-mockup deliverables.

Reference	Purpose
SC2006 Project Description	Requires meaningful use of authorised real-world data and clear user value.
FamPlan SRS Lab 1	Previous baseline from which the family-neighbourhood revision was derived.
Configured public datasets	Planned sources for schools, early-childhood services, supermarkets, healthcare, parks, public transport, maps, and routing; exact endpoints and licences must be confirmed before implementation.

1.6 Team Contributions and AI Use

Member	Username	Team Leader	Lab Contributions
POLOYAPOY RON JOSHUA ABES	POLO0002	Yes	Data Dictionary, Class Diagram
LIM RUI XIAN EDWARD	ELIM095	No	Functional and Non-functional Requirements, Class Diagram
LE VIET HOANG ANH	HOANGANH003	No	Initial Use Case Model, Class Diagram
ATHENA LIM HUI RU (LIN HUIRU)	ALIM102	No	Initial Use Case Model, Class Diagram
MUHAMMAD ILHAM BIN IRWAN	MUHA0313	No	UI Mockups, Dialog Map, Class Diagram

AI-use disclosure: OpenAI Codex assisted with consolidating team-authored artifacts, consistency editing, and populating the supplied Word templates; the team reviewed and accepts responsibility for the final content.

2. Overall Description

2.1 Product Perspective

Family-related location information is distributed across separate government and public services. FamPlan consolidates authorised records into neighbourhood profiles and applies transparent, user-controlled criteria. It stores only account, preference, shortlist, note, and synchronisation metadata required by the documented functions. Public source data is synchronised automatically and validated before a new local snapshot becomes active.

2.2 Product Functions

- Manage registration, authentication, password reset, display-name changes, and account deletion.
- Store family stages, preferred areas, commute destinations, amenity thresholds, and criterion weights.
- Search and filter neighbourhoods rather than property listings.
- Show childcare, education, groceries, healthcare, green spaces, transport, and accessibility evidence.

- Show interactive maps, distances, routes, and commute times with source and retrieval information.
- Calculate explainable family suitability scores and compare two to four neighbourhoods.
- Maintain a ten-neighbourhood shortlist with optional 500-character private notes.
- Synchronise configured public data automatically and let administrators monitor status and history.

2.3 User Classes and Characteristics

User Class	Characteristics and Permissions
Visitor	Can search and inspect neighbourhoods, family amenities, maps, routes, scores, and comparisons. Cannot save personal state.
Registered User	Inherits Visitor functions and can save family preferences, shortlist up to ten neighbourhoods, and maintain private notes.
Data Administrator	Inherits Registered User functions and can view automatic data-source status and synchronisation history. Does not manually trigger synchronisation.
Configured Public Data Source	External supporting system supplying authorised neighbourhood, facility, transport, map, or route information.

2.4 Operating Environment

FamPlan is a responsive browser-based application accessed from modern desktop and mobile browsers. A server-side application maintains accounts, preferences, cached public data, scoring logic, and automatic synchronisation. The system requires network access for current maps, routes, and source updates, while validated cached neighbourhood information remains available during supported provider outages.

2.5 Constraints Assumptions and Dependencies

ID	Assumption or Constraint	Consequence or Control
C-1	Only authorised public APIs or downloadable datasets may be used.	Confirm licences, quotas, schemas, and update frequencies before implementation.
C-2	A neighbourhood needs a documented reference point or boundary basis.	Show the basis and avoid implying that every resident is equally close to each facility.
C-3	Facility proximity does not imply admission, eligibility, vacancy, or service quality.	Display the limitation wherever education or childcare evidence is used.
C-4	Family preferences use stages and priorities, not children's names, birth dates, applications, or medical information.	Apply data minimisation in forms, storage, analytics, and logs.
C-5	Configured sources can differ in coverage and freshness.	Show category-level source, date, missing-data, and stale-data states.
C-6	Automatic source synchronisation runs at least every 24 hours.	Administrators monitor outcomes; failed candidates never replace the last validated snapshot.
A-1	OneMap or an equivalent authorised provider supports mapping, geocoding, and routing.	If coverage is unavailable, only dependent functions show an unavailable state.

ID	Assumption or Constraint	Consequence or Control
A-2	School, early-childhood, retail, healthcare, recreation, and transport sources expose sufficient location data or addresses that can be geocoded.	Unsupported categories must be revised explicitly rather than populated with invented data.

3. External Interface Requirements

3.1 User Interfaces

The interface provides labelled search and preference controls, accessible category filters, a neighbourhood summary, a map with a non-map alternative, facility lists, source and date labels, score explanations, comparison, shortlist, account pages, and an administrator status page. Education and childcare results state that proximity does not guarantee admission or availability. All screens must use the family-neighbourhood vocabulary defined in Section 6.

3.2 Hardware Interfaces

No specialised hardware is required. Users interact through a browser using a keyboard, pointer, or touch input. The application does not require device sensors or a native mobile application.

3.3 Software Interfaces

Interface	Planned Information	Use and Failure Behaviour
MOE or data.gov.sg education datasets	Primary and secondary school names, types, addresses, and published metadata	Supports school proximity; does not predict admission.
ECDA or authorised early-childhood dataset	Childcare and kindergarten locations and categories	Supports early-childhood proximity; does not claim live vacancy.
Authorised retail and healthcare datasets	Supermarket and clinic locations	Supports essential-service proximity with category-level unavailable states.
OneMap	Geocoding, map display, routing, and supported place themes	Supports maps and commute functions; outage does not disable locally cached neighbourhood data.
LTA DataMall	Public-transport stop or station information	Supplements transport accessibility evidence.
Email Delivery Service	Password-reset message delivery	Supports UC-1.3 and must not reveal whether an address is registered.

3.4 Communications Interfaces

Client-server and external-service traffic uses HTTPS with TLS 1.2 or newer. Requests use validated parameters and provider-specific timeouts and rate limits. Automatic data synchronisation is initiated by the server schedule at least every 24 hours; administrators do not manually trigger it. Password reset is the only email interface in scope.

4. Functional Requirements

The following 48 atomic functional requirements define the family-neighbourhood product. Each requirement identifies an actor, input, required output, and minimum verification evidence.

4.1 Account Management

Create and manage accounts, authentication, display names, and account deletion.

ID Priority Actor	Atomic Requirement	Input and Required Output	Verification
FR-ACC-01 Must Visitor	The system shall create a user account after receiving valid registration details and consent to the stated data use.	Input: Display name, email, password, consent Output: Account confirmation or field-specific errors	Submit valid, duplicate-email, weak-password, malformed-email, and missing-consent cases.
FR-ACC-02 Must Registered User	The system shall authenticate a Registered User with valid credentials.	Input: Email and password Output: Authenticated session or generic failure	Test valid, invalid, unknown, locked, and expired-session cases without revealing account existence.
FR-ACC-03 Must Registered User	The system shall end the current authenticated session when the user logs out.	Input: Logout action Output: Invalidated session and public-page redirect	Log out and verify that the previous session token cannot access a protected function.
FR-ACC-04 Should Registered User	The system shall support password reset through a time-limited, single-use link sent to the registered email address.	Input: Email address and new password Output: Generic confirmation and completed reset or expiry error	Test registered and unknown emails, expired and reused links, and password-policy boundaries.
FR-ACC-05 Must Registered User	The system shall allow a Registered User to update the display name stored on the account.	Input: New display name Output: Updated display name or validation error	Update the display name and verify that email and credentials cannot be changed through this function.
FR-ACC-06 Must Registered User	The system shall delete the user's account, saved preferences, shortlist, and private notes after explicit confirmation.	Input: Confirmed deletion request Output: Deletion acknowledgement and ended session	Confirm and cancel deletion, then verify that removed data is no longer retrievable within the stated period.

4.2 Family and Neighbourhood Preferences

Store family stages, preferred areas, destinations, amenity thresholds, and criterion weights.

ID Priority Actor	Atomic Requirement	Input and Required Output	Verification
FR-PREF-01 Must Registered User	The system shall store the household's selected family-planning stages, including expecting a child, infant care, preschool, primary school, and secondary school.	Input: One or more family stages Output: Saved family-stage preferences	Save every supported stage alone and in combination; reject unknown values.
FR-PREF-02 Must Registered User	The system shall store preferred towns or planning areas for neighbourhood discovery.	Input: Selected towns or planning areas Output: Saved area preferences	Save, remove, duplicate, and unknown area selections.
FR-PREF-03 Must Registered User	The system shall store up to three priority destinations, each with a travel mode, and shall use all of them for commute scoring and comparison.	Input: Destination labels, addresses and travel modes Output: Saved and geocoded destinations or field errors	Test zero through four destinations, duplicates, unresolved addresses, and every travel mode.
FR-PREF-04 Must Registered User	The system shall store selected family amenity categories and a preferred maximum distance for each selected category.	Input: Amenity categories and distance thresholds Output: Saved amenity preferences	Test all supported categories and distances below, on, and above allowed boundaries.

ID Priority Actor	Atomic Requirement	Input and Required Output	Verification
FR-PREF-05 Must Registered User	The system shall store an integer importance weight from 0 to 5 for childcare, schools, groceries, healthcare, green spaces, public transport, commute, and accessibility.	Input: Eight criterion weights Output: Saved weights or field errors	Test -1, 0, 5, 6, non-integer, partial, and all-zero weight sets.
FR-PREF-06 Must Registered User	The system shall retrieve and update the user's saved family and neighbourhood preferences.	Input: Authenticated preference request Output: Current or updated preference profile	Save preferences, sign out, sign in, retrieve them, update them, and confirm ownership isolation.
FR-PREF-07 Should Registered User	The system shall offer editable starting weights based on the selected family-planning stages without overwriting choices already confirmed by the user.	Input: Family stages and preset action Output: Editable suggested weights	Apply each preset, edit it, and verify that a later stage change does not overwrite confirmed values without consent.

4.3 Neighbourhood Discovery

Search, filter, label, sort, and validate neighbourhood profiles.

ID Priority Actor	Atomic Requirement	Input and Required Output	Verification
FR-DISC-01 Must Visitor	The system shall search neighbourhoods by town, planning area, estate name, or address.	Input: Search text or selected area Output: Matching neighbourhood profiles	Test known and unknown towns, areas, estates, addresses, spelling variants, and empty search.
FR-DISC-02 Must Visitor	The system shall filter neighbourhood results by family stage, amenity category, minimum facility count, maximum nearest-facility distance, and commute-time threshold.	Input: Filter values Output: Filtered neighbourhood profiles	Test each filter independently, in valid combinations, and at every numeric boundary.
FR-DISC-03 Must Registered User	The system shall apply saved family and neighbourhood preferences to the current search.	Input: Apply-preferences action Output: Populated filters and refreshed results	Test complete, partial, empty, and outdated preference profiles without silently changing saved values.
FR-DISC-04 Must Visitor	The system shall identify every result as a neighbourhood profile rather than a live property listing or guarantee of service availability.	Input: Search results Output: Visible neighbourhood-profile and limitation labels	Inspect every result layout, empty state, comparison, shortlist, and shared view.
FR-DISC-05 Must Visitor	The system shall sort neighbourhood results by family suitability score, nearest selected amenity, commute time, or overall amenity coverage when the selected value is available.	Input: Sort choice Output: Ordered results with unavailable values placed last	Verify ascending and descending order, ties, missing values, and stable pagination.
FR-DISC-06 Must Visitor	The system shall reject invalid or contradictory discovery criteria before executing a search.	Input: Submitted search and filter criteria Output: Field-specific correction guidance	Test negative counts, unsupported distances, invalid destinations, contradictory thresholds, and unexpected types.

4.4 Family Amenities and Neighbourhood Information

Present family-relevant facilities, proximity measures, counts, sources, and limitations.

ID Priority Actor	Atomic Requirement	Input and Required Output	Verification
FR-NBH-01 Must Visitor	The system shall display a selected neighbourhood's name, town or planning area, map location, and the boundary or coverage basis used for its analysis.	Input: Selected neighbourhood Output: Neighbourhood summary and coverage explanation	Open a known profile and verify all fields, source metadata, and missing-boundary handling.

ID Priority Actor	Atomic Requirement	Input and Required Output	Verification
FR-NBH-02 Must Visitor	The system shall retrieve nearby childcare centres, kindergartens, primary schools, secondary schools, supermarkets, clinics, parks or playgrounds, MRT stations, and bus stops for a selected neighbourhood.	Input: Neighbourhood and amenity category Output: Categorised nearby facilities or category-specific unavailable state	Use fixtures containing every category, no matches, duplicate records, and a failed provider.
FR-NBH-03 Must Visitor	The system shall allow nearby-facility results to be limited to a 500 m, 1 km, or 2 km radius.	Input: Selected radius Output: Facilities within the radius	Test places immediately inside, exactly on, and immediately outside every radius.
FR-NBH-04 Must Visitor	The system shall display the calculated distance from the neighbourhood reference point to each nearby facility and state the calculation basis.	Input: Neighbourhood and facility coordinates Output: Distance, unit, and basis	Compare displayed values with known coordinate fixtures and documented rounding.
FR-NBH-05 Must Visitor	The system shall display the number of matched facilities in each supported family amenity category for the selected radius.	Input: Categorised nearby-facility results Output: Per-category counts	Verify zero, one, many, boundary, and duplicate-source cases.
FR-NBH-06 Must Visitor	The system shall identify the nearest matched facility in each selected category when at least one match is available.	Input: Categorised facility distances Output: Nearest facility and distance or no-match state	Test ties, no matches, missing coordinates, and deterministic ordering.
FR-NBH-07 Must Visitor	The system shall display each facility's name, category, address where available, and source.	Input: Selected facility Output: Facility details with explicit unavailable fields	Inspect every category with complete and incomplete source records.
FR-NBH-08 Must Visitor	The system shall display the provider and applicable update or retrieval time for neighbourhood and facility information.	Input: External response metadata Output: Visible provenance information	Inspect fresh, cached, merged, and unavailable results.
FR-NBH-09 Must Visitor	The system shall distinguish no matched facility, incomplete category coverage, stale cached information, and source unavailability instead of presenting a misleading zero.	Input: Data availability and freshness state Output: Explicit category-level status	Simulate each state and verify that unaffected categories remain visible.
FR-NBH-10 Must Visitor	The system shall state that distance to a school or childcare centre does not guarantee admission, enrolment, vacancy, or eligibility.	Input: Education or childcare output Output: Visible limitation statement	Inspect neighbourhood, comparison, shortlist, and shared result views.

4.5 Map and Travel

Show neighbourhood maps, nearby places, routes, travel times, and supported accessibility coverage.

ID Priority Actor	Atomic Requirement	Input and Required Output	Verification
FR-TRV-01 Must Visitor	The system shall display the selected neighbourhood and matched facilities on an interactive map with distinguishable category markers.	Input: Neighbourhood and nearby facilities Output: Interactive categorised map	Test marker categories, keyboard access, overlapping points, and missing coordinates.
FR-TRV-02 Must Visitor	The system shall request a route and estimated travel time from the selected neighbourhood to a chosen priority destination.	Input: Origin, destination, and travel mode Output: Route, distance, duration, or unavailable state	Test valid, invalid, timeout, no-route, and unresolved-destination cases.
FR-TRV-03 Must Visitor	The system shall support the travel modes made available by the authorised routing provider and clearly reject unsupported modes.	Input: Travel-mode selection Output: Supported route or explanatory rejection	Test every configured and unsupported mode against controlled responses.
FR-TRV-04 Should Visitor	When barrier-free routing is requested, the system shall show a barrier-free route where the authorised provider supplies one or state that coverage is unavailable.	Input: Origin, destination, barrier-free option Output: Accessible route or coverage limitation	Test supported, unsupported, and partially covered route fixtures.

ID Priority Actor	Atomic Requirement	Input and Required Output	Verification
FR-TRV-05 Must Visitor	The system shall display the provider and retrieval time for route information.	Input: Routing response metadata Output: Visible provider and retrieval time	Inspect successful, cached, and unavailable route states.

4.6 Family Decision Support

Calculate transparent family suitability scores, compare neighbourhoods, and manage a shortlist.

ID Priority Actor	Atomic Requirement	Input and Required Output	Verification
FR-DEC-01 Must Visitor	The system shall calculate a family suitability score for each neighbourhood using the selected criterion weights and disclosed normalisation rules.	Input: Neighbourhood evidence and criterion weights Output: Total score on a 0 to 100 scale	Compare representative and boundary results with independently calculated fixtures.
FR-DEC-02 Must Visitor	The system shall display each criterion's value, normalised value, weight, and contribution so the contributions reconcile with a total rounded to the nearest whole number, with an exact 0.5 rounded upward.	Input: Score calculation Output: Itemised explanation and total	Recalculate typical, boundary, all-zero, and exact-0.5 cases.
FR-DEC-03 Must Visitor	The system shall identify every criterion lacking sufficient data and disclose how its weight was treated in the score.	Input: Neighbourhood with missing criterion data Output: Missing-data label and adjusted-weight explanation	Remove each criterion in turn and verify the disclosed rule and resulting total.
FR-DEC-04 Must Visitor	The system shall compare between two and four selected neighbourhoods in a common view.	Input: Two to four neighbourhood selections Output: Side-by-side family amenity, commute, accessibility, and score information	Test zero through five selections, duplicates, removal, and missing values.
FR-DEC-05 Must Registered User	The system shall add a selected neighbourhood to the Registered User's shortlist, up to a maximum of ten neighbourhoods, without creating a duplicate entry.	Input: Add-to-shortlist action Output: Persisted entry, already-saved state, or capacity explanation	Add through the tenth entry, attempt an eleventh, add a duplicate, and verify persistence after sign-in.
FR-DEC-06 Must Registered User	The system shall remove a selected neighbourhood from the user's shortlist.	Input: Remove action Output: Updated shortlist and confirmation	Remove existing and non-existing entries and verify persistent state.
FR-DEC-07 Should Registered User	The system shall allow a private text note of up to 500 characters to be created, edited, and cleared for each shortlisted neighbourhood.	Input: Shortlisted neighbourhood and note text Output: Persisted note or validation error	Test create, update, clear, 499-, 500-, and 501-character values, unsafe markup, and cross-account access.

4.7 Automatic Data Synchronisation and Resilience

Automatically update configured public data, let administrators monitor outcomes, and preserve validated data after failures.

ID Priority Actor	Atomic Requirement	Input and Required Output	Verification
FR-ADM-01 Must Data Administrator	The system shall require an authenticated administrator role before displaying data-source monitoring functions.	Input: Authenticated session and monitoring request Output: Monitoring page or access denial	Test anonymous, ordinary-user, expired-admin, and active-admin sessions at UI and API levels.
FR-ADM-02 Must Data Administrator	The system shall display each configured source's latest successful synchronisation time, latest attempted synchronisation time, record count, and current status.	Input: Administration-page request Output: Source-status summary	Compare the view with controlled successful, failed, never-run, and delayed states.

ID Priority Actor	Atomic Requirement	Input and Required Output	Verification
FR-ADM-03 Must System	The system shall automatically check and synchronise each configured public data source at least once every 24 hours without requiring an administrator to start the process.	Input: Configured schedule and source metadata Output: Recorded check, synchronisation outcome, and active dataset decision	Advance a test clock across the boundary and verify that no manual action is required.
FR-ADM-04 Must Data Administrator	The system shall display synchronisation history containing source, start time, end time, outcome, records added or changed, and a sanitised error summary.	Input: Source and optional date filter Output: Reverse-chronological synchronisation history	Generate successes and failures and verify fields, ordering, filtering, and absence of secrets.
FR-RES-01 Must System	The system shall preserve the last validated local dataset when an automatic synchronisation fails validation or cannot complete.	Input: Failed synchronisation Output: Previous active dataset and recorded failure	Query before and after injected download, schema, and validation failures.
FR-RES-02 Must System	The system shall isolate an external map, route, or facility-service failure so account, shortlist, and locally cached neighbourhood functions remain usable.	Input: Simulated dependency failure Output: Controlled unavailable state for affected features	Disable each dependency and execute every named unaffected workflow.
FR-RES-03 Must Visitor	The system shall show the age of cached external information and a stale-data warning when the configured freshness threshold is exceeded.	Input: Cached-result metadata Output: Data age and applicable stale warning	Test values immediately below, on, and above the configured threshold.

5. Nonfunctional Requirements

The following 29 measurable requirements constrain performance, scalability, usability, accessibility, compatibility, reliability, data quality, security, privacy, maintainability, testability, and transparency.

5.1 Performance Requirements

ID and Priority	Measurable Requirement	Verification
NFR-PERF-01 Must	The 95th-percentile server response time for neighbourhood search and filtering must not exceed 2 seconds during a 15-minute test with 100 concurrent simulated users against at least 100,000 facility and neighbourhood records, excluding live external API calls.	Load-test report containing dataset size, concurrency, request count, errors, and percentile timings.
NFR-PERF-02 Must	The 95th-percentile response time for locally calculated family scores and neighbourhood comparisons must not exceed 3 seconds under the NFR-PERF-01 workload.	Load-test representative two-, three-, and four-neighbourhood comparisons.
NFR-PERF-03 Should	The 95th-percentile end-to-end response time for a live route request must not exceed 5 seconds while the provider is within documented normal operating conditions.	Timed integration tests that distinguish provider time from local processing and include controlled timeouts.

5.2 Scalability Requirements

ID and Priority	Measurable Requirement	Verification
NFR-SCALE-01 Must	The deployed application must store, index, and query at least 100,000 neighbourhood, facility, transport, and provenance records without violating NFR-PERF-01 or NFR-PERF-02.	Seed the required volume, record indexes, and execute the performance tests.
NFR-SCALE-02 Must	The deployed application must support at least 100 concurrent active users performing representative search, profile, map, score, comparison, shortlist, and cached-route workflows without violating applicable performance targets.	Mixed-workload load test with response and error statistics by operation.

5.3 Usability Requirements

ID and Priority	Measurable Requirement	Verification
NFR-USAB-01 Should	At least 80% of first-time representative users in a moderated test of at least 10 participants must search for a neighbourhood, select two results, and open their comparison within 3 minutes without facilitator assistance.	Anonymised usability protocol, observations, timings, completion rate, and issues.
NFR-USAB-02 Must	Every rejected user input must identify the affected field or action and state how the user can correct it; a generic error code alone is insufficient.	UI and API acceptance tests covering every documented validation error.

5.4 Accessibility Requirements

ID and Priority	Measurable Requirement	Verification
NFR-ACC-01 Must	The account, preference, search, neighbourhood, map, comparison, shortlist, and administration workflows must conform to WCAG 2.1 Level AA.	Automated accessibility scan plus manual checks for labels, headings, contrast, errors, zoom, maps, and screen-reader output.
NFR-ACC-02 Must	Every interactive function in the core workflows must be operable using only a keyboard and must expose a visible focus indicator.	Manual keyboard traversal using a documented checklist with no pointer input.

5.5 Compatibility Requirements

ID and Priority	Measurable Requirement	Verification
NFR-COMP-01 Must	Core pages must remain usable from 360 px to 1440 px viewport widths without horizontal scrolling of the primary page content.	Screenshot or browser-test evidence at boundary and intermediate widths.
NFR-COMP-02 Should	Core workflows must operate on the latest two stable versions available at test time of Chrome, Safari, Firefox, and Edge.	Compatibility matrix recording browser versions and pass or fail results.

5.6 Reliability Requirements

ID and Priority	Measurable Requirement	Verification
NFR-REL-01 Must	At least 99% of core requests must complete successfully during a continuous one-hour representative-load test, excluding requests whose sole failure is an injected or confirmed external-provider outage.	One-hour report with request totals, failure classification, and evidence for exclusions.

5.7 Recoverability Requirements

ID and Priority	Measurable Requirement	Verification
NFR-REL-02 Must	Following an application restart, locally stored account, preference, search, profile, shortlist, administration, and cached-neighbourhood functions must become operational within 5 minutes without manual reconstruction of persistent data.	Timed restart-and-recovery exercise followed by integrity and smoke tests.

5.8 Fault isolation Requirements

ID and Priority	Measurable Requirement	Verification

ID and Priority	Measurable Requirement	Verification
NFR-REL-03 Must	Failure of one external dependency must not prevent use of unrelated functions backed by validated local data.	Dependency-failure matrix showing affected and unaffected workflows.

5.9 Data provenance Requirements

ID and Priority	Measurable Requirement	Verification
NFR-DATA-01 Must	Every neighbourhood, facility, transport, route, and family-score output must display or link to its source and applicable update or retrieval date.	Automated presence checks plus manual inspection of every output and error state.

5.10 Data freshness Requirements

ID and Priority	Measurable Requirement	Verification
NFR-DATA-02 Must	Locally cached neighbourhood and facility data older than 31 complete days must be visibly marked as stale until a newer validated dataset becomes active.	Set test-data ages to 30, 31, and 32 complete days and verify boundary behaviour.

5.11 Reproducibility Requirements

ID and Priority	Measurable Requirement	Verification
NFR-DATA-03 Must	Identical neighbourhood inputs, user weights, scoring-configuration version, and dataset versions must produce identical family scores and comparison results.	Repeat deterministic fixture tests across process restarts.

5.12 Data quality Requirements

ID and Priority	Measurable Requirement	Verification
NFR-DATA-04 Must	Facility records merged from multiple sources must be normalised by supported category and must not be counted twice when they represent the same named facility at the same location.	Use exact, near-duplicate, renamed, and distinct co-located facility fixtures and inspect counts.

5.13 Transport security Requirements

ID and Priority	Measurable Requirement	Verification
NFR-SEC-01 Must	Every production client-server connection must use HTTPS with TLS 1.2 or newer, and plain HTTP must redirect to HTTPS without accepting credentials.	Protocol scan, redirect test, and attempted authentication over plain HTTP.

5.14 Credential security Requirements

ID and Priority	Measurable Requirement	Verification
NFR-SEC-02 Must	Passwords must be stored only as salted hashes using a current approved password-hashing algorithm and must never appear in application logs.	Inspect schema and configuration; scan logs after registration, login, failure, and reset tests.

5.15 Authorisation Requirements

ID and Priority	Measurable Requirement	Verification
NFR-SEC-03 Must	A non-administrator must be unable to read data-source monitoring functions even when directly calling their endpoints.	Role-based access-control tests at both UI and API levels.

5.16 Authentication security Requirements

ID and Priority	Measurable Requirement	Verification
NFR-SEC-04 Must	An account must be locked from password authentication for 15 minutes after five consecutive failed login attempts; a successful login before the threshold must reset the counter.	Test attempts 1-6, successful reset, and 14-, 15-, and 16-minute boundaries.

5.17 Input security Requirements

ID and Priority	Measurable Requirement	Verification
NFR-SEC-05 Must	All user-controlled input must be allow-list or structurally validated before use in a query, calculation, stored note, map request, or external-service request.	Security tests for injection, unsafe markup, malformed coordinates, invalid ranges, and unexpected types.

5.18 Privacy Requirements

ID and Priority	Measurable Requirement	Verification
NFR-PRIV-01 Must	Account details, saved family preferences, shortlists, and notes must become irretrievable through the application within 24 hours after confirmed account deletion.	Delete a seeded account and test UI, API, and authorised data-store lookup at the deadline.

5.19 Data minimisation Requirements

ID and Priority	Measurable Requirement	Verification
NFR-PRIV-02 Must	Registration and family preferences must not require exact dates of birth, children's names, school applications, medical information, or other personal information unnecessary for the documented functions.	Compare required fields with the approved data dictionary and submit all workflows without excluded fields.

5.20 Maintainability Requirements

ID and Priority	Measurable Requirement	Verification
NFR-MAINT-01 Must	Data-source endpoints, freshness thresholds, amenity categories, scoring criteria, and normalisation constants must be maintained as versioned configuration or isolated modules so one category can be updated without changing unrelated workflow code.	Code review plus tests that replace one configuration category while unaffected workflows still pass.

5.21 Testability Requirements

ID and Priority	Measurable Requirement	Verification

ID and Priority	Measurable Requirement	Verification
NFR-MAINT-02 Should	Automated tests must achieve at least 80% branch coverage for family scoring, proximity aggregation, facility deduplication, and external-data transformation modules.	Coverage report generated by the project's documented test command.

5.22 Transparency Requirements

ID and Priority	Measurable Requirement	Verification
NFR-TRANS-01 Must	Every family suitability score must expose criterion values, weights, contributions, dataset versions, and missing-data treatment needed to reproduce it subject to documented rounding.	Recalculate representative outputs independently from the displayed explanation.

5.23 Responsible guidance Requirements

ID and Priority	Measurable Requirement	Verification
NFR-TRANS-02 Must	The user interface must state that neighbourhood scores are comparative planning aids and do not guarantee school admission, childcare availability, enrolment eligibility, property availability, or suitability for every family.	Review every score, comparison, shortlist, and shareable output for the limitation statement.

5.24 Business Rules

Rule	Value	Requirements
Neighbourhood comparison	2 to 4 neighbourhoods	FR-DEC-04
Shortlist capacity	Maximum 10 neighbourhoods per Registered User	FR-DEC-05
Private note length	0 to 500 characters	FR-DEC-07
Proximity radii	500 m, 1 km, or 2 km	FR-NBH-03
Criterion weights	Integer 0 to 5	FR-PREF-05
Score rounding	Nearest whole number; exact 0.5 rounds upward	FR-DEC-02
Automatic synchronisation	At least once every 24 hours; system initiated	FR-ADM-03
Editable account detail	Display name	FR-ACC-05
Commute time	Average across all destinations, each by its travel mode	FR-PREF-03; FR-DEC-01

6. Data Dictionary

The table defines the shared terms used by the requirements, use cases, implementation, and future tests. Examples are illustrative; the attributes and constraints are the baseline.

Term	Definition	Key Attributes	Relationships and Constraints
User Account	Registered person's identity in FamPlan.	id; display_name; email; consent_given_at; status	Owns one preference profile and zero or more shortlist entries; display name is the editable general account field.
Password Credential	Stored proof of a password, never the password itself.	account_id; salted_hash; algorithm; parameters	Belongs to one User Account; never logged.

Term	Definition	Key Attributes	Relationships and Constraints
Session	Current authenticated login.	token; account_id; created_at; expires_at; invalidated_at	Belongs to one User Account; invalidated immediately on logout.
Password Reset Token	Time-limited, single-use credential for account recovery.	token; account_id; expires_at; used_at	Belongs to one User Account; delivery must not reveal whether an email exists.
Administrator Role	Privilege permitting data-source monitoring.	account_id; granted_at	At most one per User Account; enforced at UI and API levels.
Family Preference Profile	Saved settings used to personalise neighbourhood discovery.	account_id; family_stages; preferred_areas; amenity_thresholds; criterion_weights	At most one per Registered User; applied to a Search Query.
Family Stage	Planning stage relevant to nearby services.	enum: expecting_child; infant_care; preschool; primary_school; secondary_school	One or more values may be selected; no names or dates of birth are required.
Amenity Category	Supported type of family-relevant place.	childcare; kindergarten; primary_school; secondary_school; supermarket; clinic; park_playground; mrt; bus_stop	Classifies Facility records and links preferences to score criteria.
Criterion Weight	User importance assigned to a scoring criterion.	criterion; integer weight 0 to 5	Part of a Family Preference Profile; all-zero input uses the documented fallback.
Search Query	Request-scoped neighbourhood search and filters.	search_text; area; stages; categories; counts; distances; commute threshold; sort	May be populated from saved preferences; invalid combinations are rejected before execution.
Neighbourhood Profile	Comparable representation of a town, planning area, estate, or documented coverage zone.	id; name; town; planning_area; reference_point; boundary_basis	Has many Facility Matches, routes, score breakdowns, and shortlist references; it is not a property listing.
Geolocation	Latitude and longitude pair.	latitude; longitude; source	Attached to neighbourhoods, facilities, and destinations.
Facility	Childcare, education, retail, healthcare, recreation, or transport place.	id; name; category; address; geolocation; source_id; source_updated_at	May appear in many neighbourhood-radius results; duplicates are resolved by the documented matching rule.
Proximity Radius	Boundary used when retrieving nearby facilities.	metres: 500; 1000; 2000	Bounds Facility Matches for one neighbourhood query.
Facility Match	Facility located within a selected radius of a neighbourhood reference point.	neighbourhood_id; facility_id; distance_m; distance_basis	Joins one Neighbourhood Profile to one Facility for a specific calculation.
Amenity Summary	Per-category neighbourhood aggregation.	category; matched_count; nearest_facility_id; nearest_distance_m; availability_state	Computed from Facility Matches and used by scoring and comparison.
Priority Destination	Saved place and travel mode used for commute scoring and comparison.	slot 1 to 3; label; query_text; geolocation; travel_mode	Part of a Family Preference Profile and destination of a Route Estimate.
Route Estimate	Provider response for travel from a neighbourhood to a destination.	origin; destination; mode; distance_m; duration_s; retrieved_at; state	Carries provenance and may include a barrier-free coverage flag.
Scoring Configuration	Versioned definitions and normalisation constants for family suitability.	version; effective_date; criteria; normalisation rules	Referenced by every Family Suitability Score.
Family Suitability Score	0-100 indication of how well a neighbourhood matches selected priorities.	neighbourhood_id; configuration_version; dataset_versions; total; computed_at	Contains Criterion Score Breakdowns; comparative planning aid, not a guarantee.
Criterion Score Breakdown	One criterion's transparent contribution to a total score.	criterion; raw_value; normalised_value; weight; contribution	Part of one Family Suitability Score.

Term	Definition	Key Attributes	Relationships and Constraints
Missing Data Treatment	Explanation of how unavailable evidence affects scoring.	criterion; reason; weight_treatment	Attached to a score breakdown when a value cannot be computed.
Neighbourhood Comparison Set	Transient selection of two to four neighbourhoods.	neighbourhood_ids	Request-scoped; contains no duplicates.
Shortlist Entry	Saved neighbourhood selected by a Registered User.	account_id; neighbourhood_id; added_at	Maximum ten per account; unique by account and neighbourhood; may have one note.
Shortlist Note	Private text attached to a shortlisted neighbourhood.	shortlist_entry_id; text; updated_at	Maximum 500 characters; readable only by the owning account.
Data Source Registry	Configuration and current health of each public data source.	source_id; name; schedule; freshness_threshold; last_attempt; last_success; status	Has many Sync History Records and Validated Dataset Snapshots.
Sync History Record	Outcome of one automatic API update attempt.	source_id; start; end; outcome; records_added; records_changed; sanitised_error	Created automatically; displayed to the Data Administrator; contains no manual-trigger field.
Validated Dataset Snapshot	Source version that passed validation and can be served.	source_id; version; validated_at; active	Only a validated snapshot becomes active; prior active data remains after failure.
Data Provenance Stamp	Source and applicable data date shown with externally derived output.	provider; source_updated_at; retrieved_at; dataset_version	Attached to neighbourhood, facility, route, and score evidence.

7. Use Case Model

7.1 Manual UML Diagram Blueprint

The team will draw the UML diagram manually. Use one system boundary labelled FamPlan and place the 16 use-case ellipses inside it. Keep actors outside the boundary and draw each solid association independently from an actor to a use case. Do not connect actors by drawing solid lines through multiple use cases.

Actor Label	Placement	Generalisation or Meaning
Visitor	Left of the FamPlan boundary	Base public actor.
Registered User	Below Visitor	Draw a hollow-triangle generalisation arrow from Registered User to Visitor.
Data Administrator	Below Registered User	Draw a hollow-triangle generalisation arrow from Data Administrator to Registered User.
Email Delivery Service	Right of the account group	Supporting actor associated only with UC-1.3 Reset Password.
OneMap	Right of the map and travel group	Supporting actor associated with UC-5.1 and UC-5.2.
Configured Public Data Sources	Right of the amenity and administration groups	Supporting actor associated with UC-4.2, UC-5.1, and UC-7.1.

7.2 System Boundary Groupings

Group Label	Use Cases to Place Inside

Group Label	Use Cases to Place Inside
1 Account Management	UC-1.1 Register Account; UC-1.2 Login and Logout; UC-1.3 Reset Password; UC-1.4 Manage Account Details; UC-1.5 Delete Account
2 Family Preferences	UC-2.1 Manage Family Preferences
3 Neighbourhood Discovery	UC-3.1 Search Neighbourhoods; UC-3.2 Apply Saved Preferences
4 Neighbourhood and Family Amenities	UC-4.1 View Neighbourhood Profile; UC-4.2 Explore Family Amenities
5 Map and Travel	UC-5.1 View Map and Nearby Places; UC-5.2 Check Commute Route
6 Decision Support	UC-6.1 Calculate Family Suitability Score; UC-6.2 Compare Neighbourhoods; UC-6.3 Manage Shortlist and Notes
7 Data Monitoring	UC-7.1 Monitor Automatic Data Synchronisation

7.3 Actor Associations

Actor	Draw Solid Association Lines To
Visitor	UC-1.1, UC-3.1, UC-4.1, UC-4.2, UC-5.1, UC-5.2, and UC-6.2
Registered User	UC-1.2, UC-1.3, UC-1.4, UC-1.5, UC-2.1, UC-3.2, and UC-6.3; inherited Visitor associations do not need to be redrawn
Data Administrator	UC-7.1 only; inherited Registered User and Visitor associations do not need to be redrawn
Email Delivery Service	UC-1.3
OneMap	UC-5.1 and UC-5.2
Configured Public Data Sources	UC-4.2, UC-5.1, and UC-7.1

7.4 UML Relationship Labels

Source	Target	Label and Arrow Direction
UC-3.2 Apply Saved Preferences	UC-3.1 Search Neighbourhoods	Dashed arrow from UC-3.2 to UC-3.1 labelled <<extend>>
UC-4.1 View Neighbourhood Profile	UC-6.1 Calculate Family Suitability Score	Dashed arrow from UC-4.1 to UC-6.1 labelled <<include>>
UC-6.2 Compare Neighbourhoods	UC-6.1 Calculate Family Suitability Score	Dashed arrow from UC-6.2 to UC-6.1 labelled <<include>>

Do not give UC-6.1 a direct actor association because it is included behaviour. Connect UC-7.1 directly to Data Administrator because the administrator only monitors the automatic process. Represent the actual scheduled API synchronisation as supporting system behaviour under FR-ADM-03, not as an administrator-initiated use case.

7.5 Use Case Catalogue

ID	Use Case	Actor or Supporting System	Goal	Requirements	Relationship
UC-1.1	Register Account	Visitor	Create a personal account.	FR-ACC-01	None
UC-1.2	Login and Logout	Registered User	Start or end an authenticated session.	FR-ACC-02 to FR-ACC-03	None

ID	Use Case	Actor or Supporting System	Goal	Requirements	Relationship
UC-1.3	Reset Password	Registered User; Email Delivery Service	Recover access through a secure reset link.	FR-ACC-04	None
UC-1.4	Manage Account Details	Registered User	Update the account display name.	FR-ACC-05	None
UC-1.5	Delete Account	Registered User	Confirm deletion of the account and saved data.	FR-ACC-06	None
UC-2.1	Manage Family Preferences	Registered User	Save family stages, areas, amenities, destinations, and weights.	FR-PREF-01 to FR-PREF-07	None
UC-3.1	Search Neighbourhoods	Visitor	Find and filter neighbourhood profiles.	FR-DISC-01 to FR-DISC-02; FR-DISC-04 to FR-DISC-06	Base use case
UC-3.2	Apply Saved Preferences	Registered User	Populate search criteria from saved preferences.	FR-DISC-03	Extends UC-3.1
UC-4.1	View Neighbourhood Profile	Visitor	Inspect the selected neighbourhood and its family score.	FR-NBH-01; FR-DEC-01 to FR-DEC-03	Includes UC-6.1
UC-4.2	Explore Family Amenities	Visitor; Configured Public Data Sources	Review nearby childcare, schools, groceries, healthcare, recreation, and transport.	FR-NBH-02 to FR-NBH-10	None
UC-5.1	View Map and Nearby Places	Visitor; Configured Public Data Sources	Explore categorised facilities and distances on a map.	FR-NBH-02 to FR-NBH-09; FR-TRV-01	None
UC-5.2	Check Commute Route	Visitor; OneMap	Request a route and travel estimate to a priority destination.	FR-TRV-02 to FR-TRV-05	None
UC-6.1	Calculate Family Suitability Score	No initiating actor	Compute an explainable neighbourhood score.	FR-DEC-01 to FR-DEC-03	Included by UC-4.1 and UC-6.2
UC-6.2	Compare Neighbourhoods	Visitor	Compare two to four neighbourhoods consistently.	FR-DEC-04; FR-DEC-01 to FR-DEC-03	Includes UC-6.1
UC-6.3	Manage Shortlist and Notes	Registered User	Save up to ten neighbourhoods and maintain private notes.	FR-DEC-05 to FR-DEC-07	None
UC-7.1	Monitor Automatic Data Synchronisation	Data Administrator; Configured Public Data Sources	Review status and history of automatic API updates.	FR-ADM-01 to FR-ADM-04	Monitoring only; synchronisation is automatic

7.6 Functional Requirement Traceability

FR ID	Use Case or Binding	Coverage
FR-ACC-01	UC-1.1	Directly covered
FR-ACC-02	UC-1.2	Directly covered
FR-ACC-03	UC-1.2	Directly covered
FR-ACC-04	UC-1.3	Directly covered
FR-ACC-05	UC-1.4	Directly covered
FR-ACC-06	UC-1.5	Directly covered
FR-PREF-01	UC-2.1	Directly covered
FR-PREF-02	UC-2.1	Directly covered
FR-PREF-03	UC-2.1	Directly covered
FR-PREF-04	UC-2.1	Directly covered
FR-PREF-05	UC-2.1	Directly covered
FR-PREF-06	UC-2.1	Directly covered

FR ID	Use Case or Binding	Coverage
FR-PREF-07	UC-2.1	Directly covered
FR-DISC-01	UC-3.1	Directly covered
FR-DISC-02	UC-3.1	Directly covered
FR-DISC-03	UC-3.2	Directly covered
FR-DISC-04	UC-3.1	Directly covered
FR-DISC-05	UC-3.1	Directly covered
FR-DISC-06	UC-3.1	Directly covered
FR-NBH-01	UC-4.1	Directly covered
FR-NBH-02	UC-4.2, UC-5.1	Directly covered
FR-NBH-03	UC-4.2, UC-5.1	Directly covered
FR-NBH-04	UC-4.2, UC-5.1	Directly covered
FR-NBH-05	UC-4.2, UC-5.1	Directly covered
FR-NBH-06	UC-4.2, UC-5.1	Directly covered
FR-NBH-07	UC-4.2, UC-5.1	Directly covered
FR-NBH-08	UC-4.2, UC-5.1	Directly covered
FR-NBH-09	UC-4.2, UC-5.1	Directly covered
FR-NBH-10	UC-4.2, UC-5.1	Directly covered
FR-TRV-01	UC-5.1	Directly covered
FR-TRV-02	UC-5.2	Directly covered
FR-TRV-03	UC-5.2	Directly covered
FR-TRV-04	UC-5.2	Directly covered
FR-TRV-05	UC-5.2	Directly covered
FR-DEC-01	UC-6.1	Directly covered
FR-DEC-02	UC-6.1	Directly covered
FR-DEC-03	UC-6.1	Directly covered
FR-DEC-04	UC-6.2	Directly covered
FR-DEC-05	UC-6.3	Directly covered
FR-DEC-06	UC-6.3	Directly covered
FR-DEC-07	UC-6.3	Directly covered
FR-ADM-01	UC-7.1	Directly covered
FR-ADM-02	UC-7.1	Directly covered
FR-ADM-03	Supporting system behaviour	Supporting or cross-cutting behaviour
FR-ADM-04	UC-7.1	Directly covered
FR-RES-01	Supporting system behaviour	Supporting or cross-cutting behaviour
FR-RES-02	Cross-cutting	Supporting or cross-cutting behaviour
FR-RES-03	Cross-cutting	Supporting or cross-cutting behaviour

7.7 Recommended UI Mockup Set

Screen	Purpose	Primary Use Cases
Neighbourhood discovery	Search, filters, saved-preference action, and result cards	UC-3.1 and UC-3.2
Family preference setup	Family stages, amenity distances, destinations, and criterion weights	UC-2.1
Neighbourhood profile	Summary, category counts, nearest facilities, sources, and score	UC-4.1, UC-4.2, and UC-6.1
Map and commute	Categorised facilities, radii, route, distance, duration, and availability	UC-5.1 and UC-5.2
Comparison and shortlist	Two-to-four comparison, score breakdowns, save action, and notes	UC-6.2 and UC-6.3
Automatic data status	Source status, last attempt, last success, counts, errors, and automatic update history	UC-7.1

8. Technology Stack Recommendation

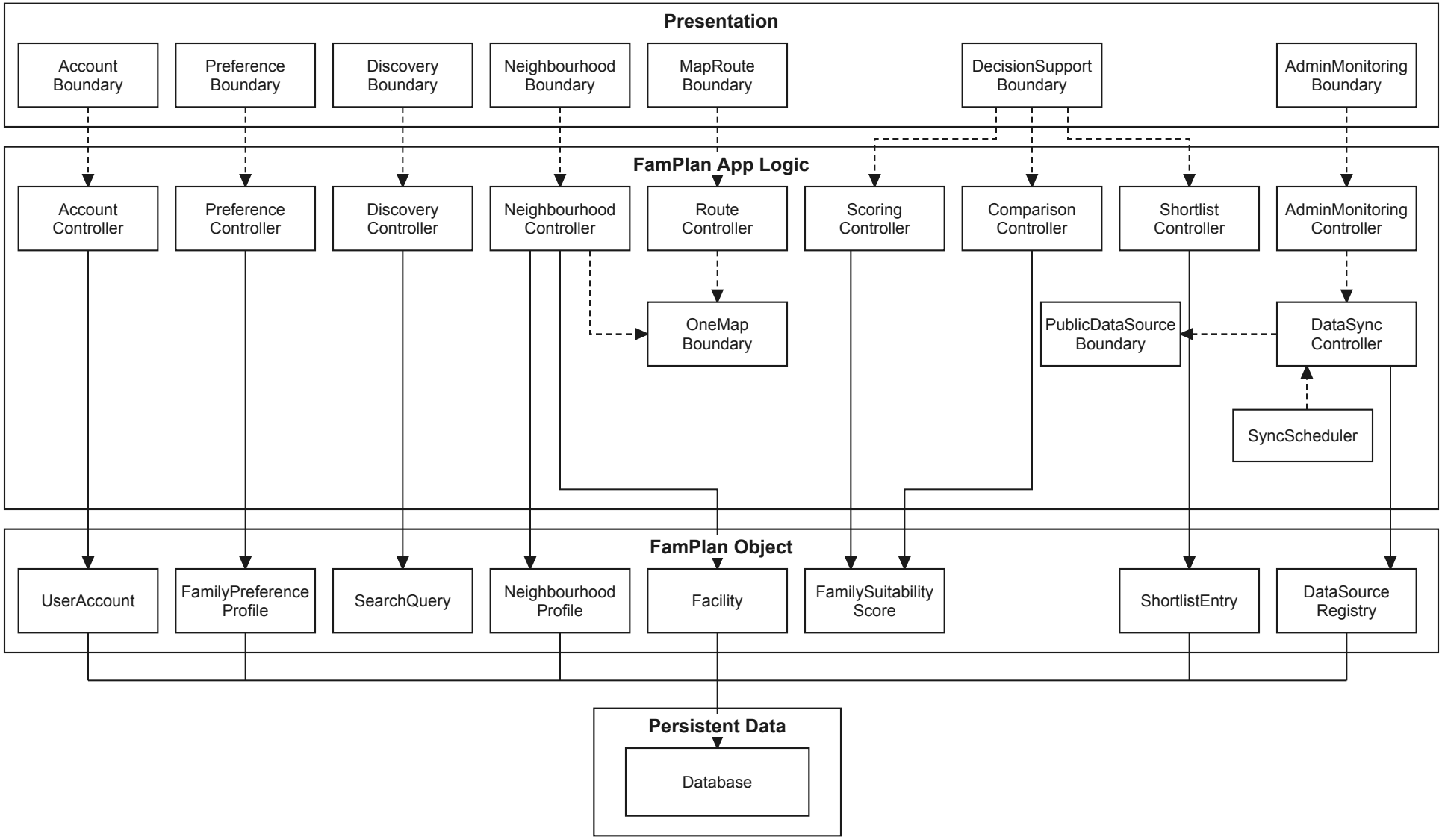
This is a deliberately small recommendation rather than a binding requirement. It keeps the project understandable for the team and supports the client-server, mapping, scoring, persistence, testing, and automatic synchronisation work already required.

Layer	Recommendation	Reason
Frontend	React with TypeScript and Vite	One responsive browser interface with reusable components and strong typing.
Backend	Node.js with Express and TypeScript	Simple REST API and one language across client and server.
Database	PostgreSQL with Prisma	Relational data, geolocation fields, constraints, migrations, and straightforward querying.
Maps and routes	Leaflet with authorised OneMap services	Lightweight interactive maps without introducing a separate native application.
Automatic data sync	A scheduled job inside the backend using node-cron	Calls configured APIs at least daily, validates data, and records outcomes without an administrator trigger.
Testing	Vitest, Supertest, and Playwright	Covers calculation units, API behaviour, and a small number of end-to-end workflows.
Local setup and deployment	GitHub, Vercel and Neon PostgreSQL	Keeps development reproducible without microservices, Kubernetes, or extra infrastructure.

8.1 Architecture Boundary

Use one frontend, one backend, and one PostgreSQL database. Keep data-source adapters and scoring logic as modules inside the backend rather than separate services. The scheduled job calls the same validation and activation modules used by the application. Do not introduce microservices, message queues, machine learning, a native mobile application, or a separate administrator backend for the first release.

8.2 System Architecture



Layered architecture: dashed arrows are calls and dependencies; solid arrows show the entity each controller reads or updates; persistent entities are stored in PostgreSQL.
Deployment: the React client is served by Vercel; Express runs as serverless Node.js functions in Singapore (sin1); Prisma connects to hosted PostgreSQL; one Vercel Cron job refreshes each of the 9 public data sources daily. Local development runs the same Express app as one long-running server with node-cron.

Presentation Layer

This layer is responsible for user interaction and view rendering inside the user's web browser, implemented as a TypeScript Single-Page Application using React and Vite. The boundary components capture user inputs, enforce form formatting, and delegate actions to backend REST API endpoints.

This layer consists of:

1. **AccountBoundary**: Presents registration, login, logout, display name updates, and password reset forms. Captures credentials and displays operation outcomes to users.
2. **PreferenceBoundary**: Captures family life stages, preferred planning areas, priority commute destinations with travel modes, and 0–5 criterion importance weights.
3. **DiscoveryBoundary**: Captures neighbourhood search queries and filter options (amenity distances, counts, and commute limits). Presents filterable and sortable result cards.
4. **NeighbourhoodBoundary**: Displays neighbourhood summary profiles, amenity counts by category, and availability states.
5. **MapRouteBoundary**: Renders an interactive Leaflet map marking categorized facilities, proximity radii (500 m, 1 km, 2 km), and travel route estimates.
6. **DecisionSupportBoundary**: Presents suitability score breakdowns, captures comparison selections (2 to 4 neighbourhoods), and displays the user's shortlisted neighbourhoods and private notes.
7. **AdminMonitoringBoundary**: Displays the automated synchronisation dashboard showing data source health, record counts, execution timestamps, and sync logs.

App Logic Layer

This layer contains all controller classes and external adapter boundaries executing on the Node.js / Express backend server. It coordinates business rules, processes data transformations, and encapsulates third-party integrations.

This layer consists of:

1. **AccountController**: Enforces password hashing using Argon2id, verifies session tokens, enforces login lockout policies (15 minutes after 5 failures), and manages account deletion lifecycles.

2. **PreferenceController**: Evaluates family planning stages, supplies stage-based weight presets, and manages persistent user preference profiles.
3. **DiscoveryController**: Validates search query bounds and executes spatial and attribute filters over cached neighbourhood datasets.
4. **NeighbourhoodController**: Aggregates nearby facilities within requested radius buffers and compiles the amenity summary for neighbourhood profiles.
5. **RouteController**: Validates transit modes and calculates commute estimates between neighbourhood reference points and priority destinations.
6. **ScoringController**: Implements the deterministic scoring engine, normalizes 8 criterion values, applies missing-data policies, and computes the rounded 0–100 suitability score.
7. **ComparisonController**: Validates that 2 to 4 neighbourhoods are selected and compiles side-by-side metric evaluations.
8. **ShortlistController**: Enforces the 10-entry shortlist capacity limit per user and validates private text notes up to 500 characters.
9. **AdminMonitoringController**: Enforces administrator role authorization and retrieves data source health and synchronisation audit logs.
10. **DataSyncController**: Fetches datasets from external endpoints, validates dataset schemas against checksums, creates provenance stamps, and persists validated snapshots.
11. **SyncScheduler**: Runs scheduled data synchronisation at least once every 24 hours without human intervention. It uses node-cron on a long-running server and one Vercel Cron job per source in the serverless production deployment.
12. **OneMapBoundary**: External adapter interfacing with the Singapore OneMap REST API for address geocoding, reverse geocoding, and transit routing.
13. **PublicDataSourceBoundary**: External adapter encapsulating public data providers (ECDA childcare, MOE schools, URA subzones, LTA DataMall) to insulate domain logic against schema changes.

App Object Layer

This layer consists of the core domain entity classes depicted in the architecture diagram:

1. **UserAccount**: Represents a registered user identity and encapsulates account status, timestamps, and role privileges.
2. **FamilyPreferenceProfile**: Stores user-configured family planning stages, preferred planning areas, and amenity distance criteria.
3. **SearchQuery**: Encapsulates transient request criteria, location text, and active filter thresholds during neighbourhood discovery.
4. **NeighbourhoodProfile**: Represents a comparable planning zone with name, planning area, reference point coordinates, and boundary basis.
5. **Facility**: Represents an individual school, clinic, preschool, supermarket, park, or transport station with geocoordinates and source metadata.
6. **FamilySuitabilityScore**: Encapsulates the overall 0–100 score, calculation configuration version, and contributing criterion breakdowns.
7. **ShortlistEntry**: Represents a saved neighbourhood pinned to a user account, supporting an optional private note.
8. **DataSourceRegistry**: Maintains configuration, freshness thresholds, and operational health status for each public data source.

Persistent Data Layer

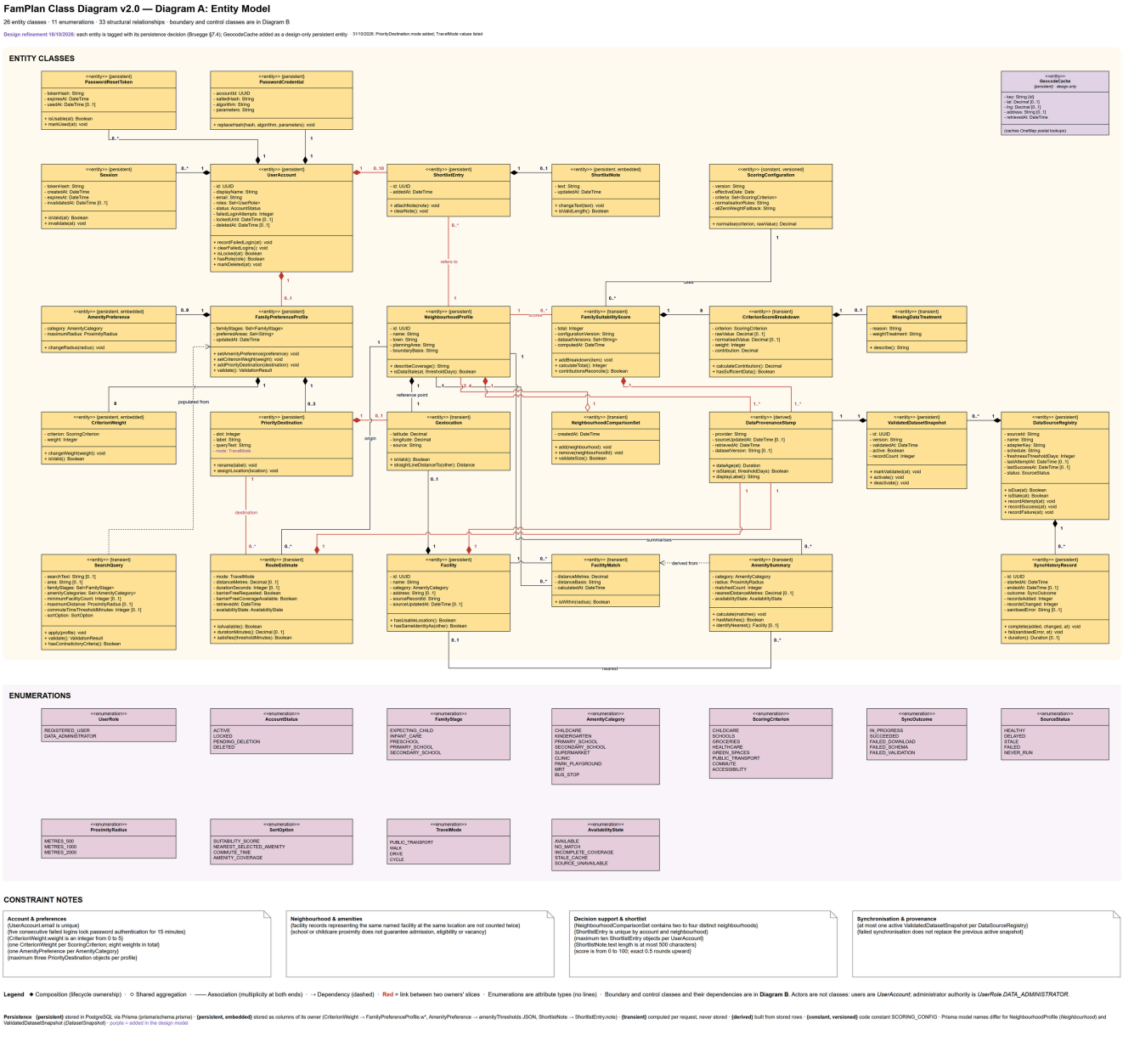
This layer contains the database that will store all of the entities. In FamPlan, this is implemented using a PostgreSQL relational database to reliably manage and persist application data.

9. Class Diagram

If the images are unclear, please refer to the raw files uploaded together with this document.

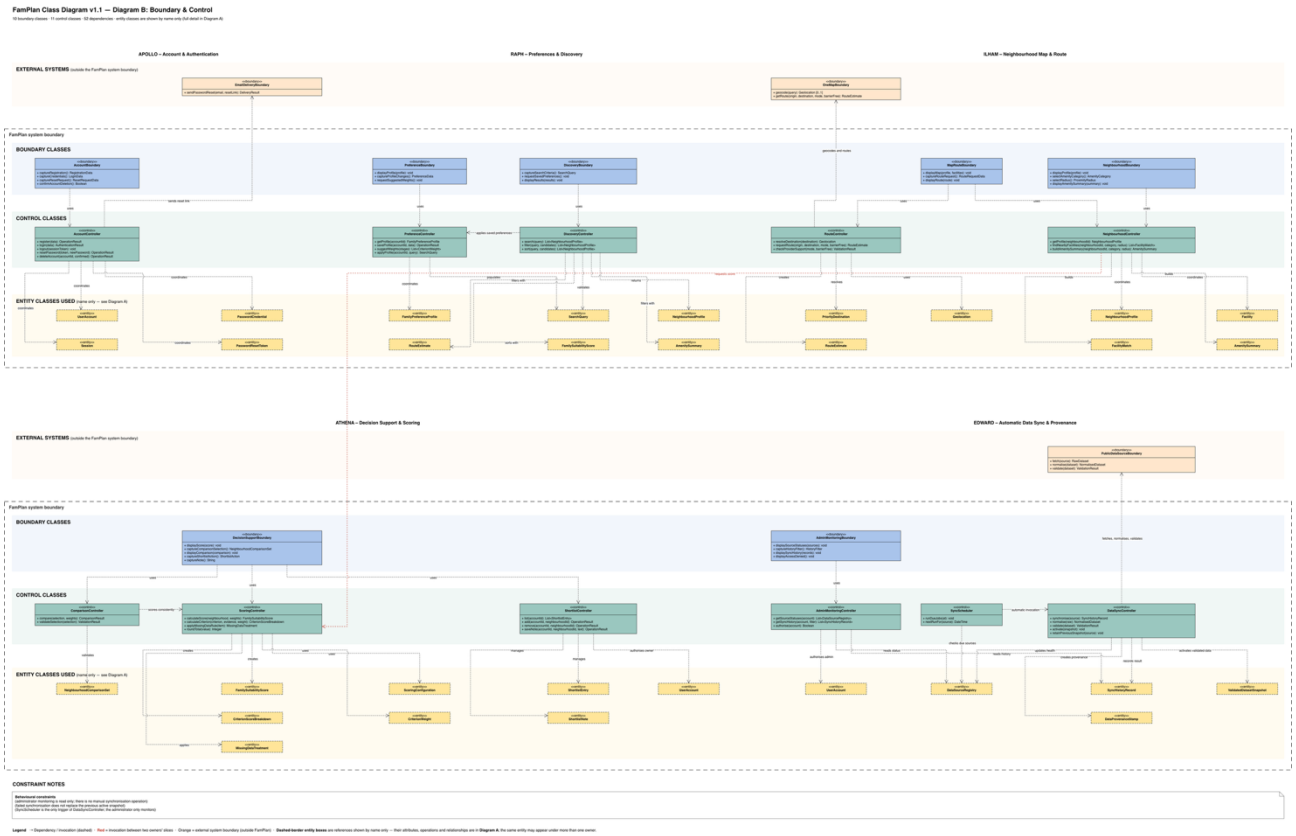
9.1 Entity Model

Domain data model defining 26 entity classes, attributes, relationships, and 11 enumerations implemented via Prisma/PostgreSQL. Each entity is tagged with its persistence decision ({persistent}, {transient}, ...); GeocodeCache is a design-only persistent class.



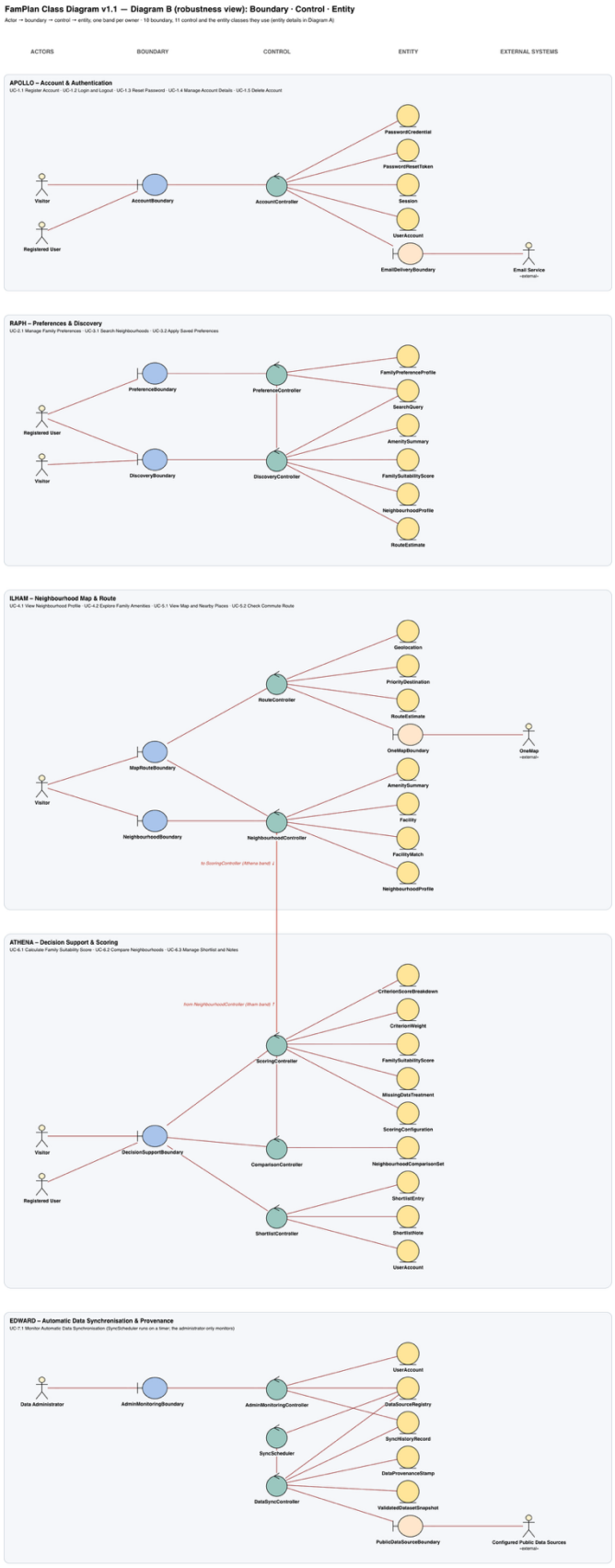
9.2 Boundary and Control Subsystems

Structural view mapping UI boundaries to controllers across the 5 functional subsystems.



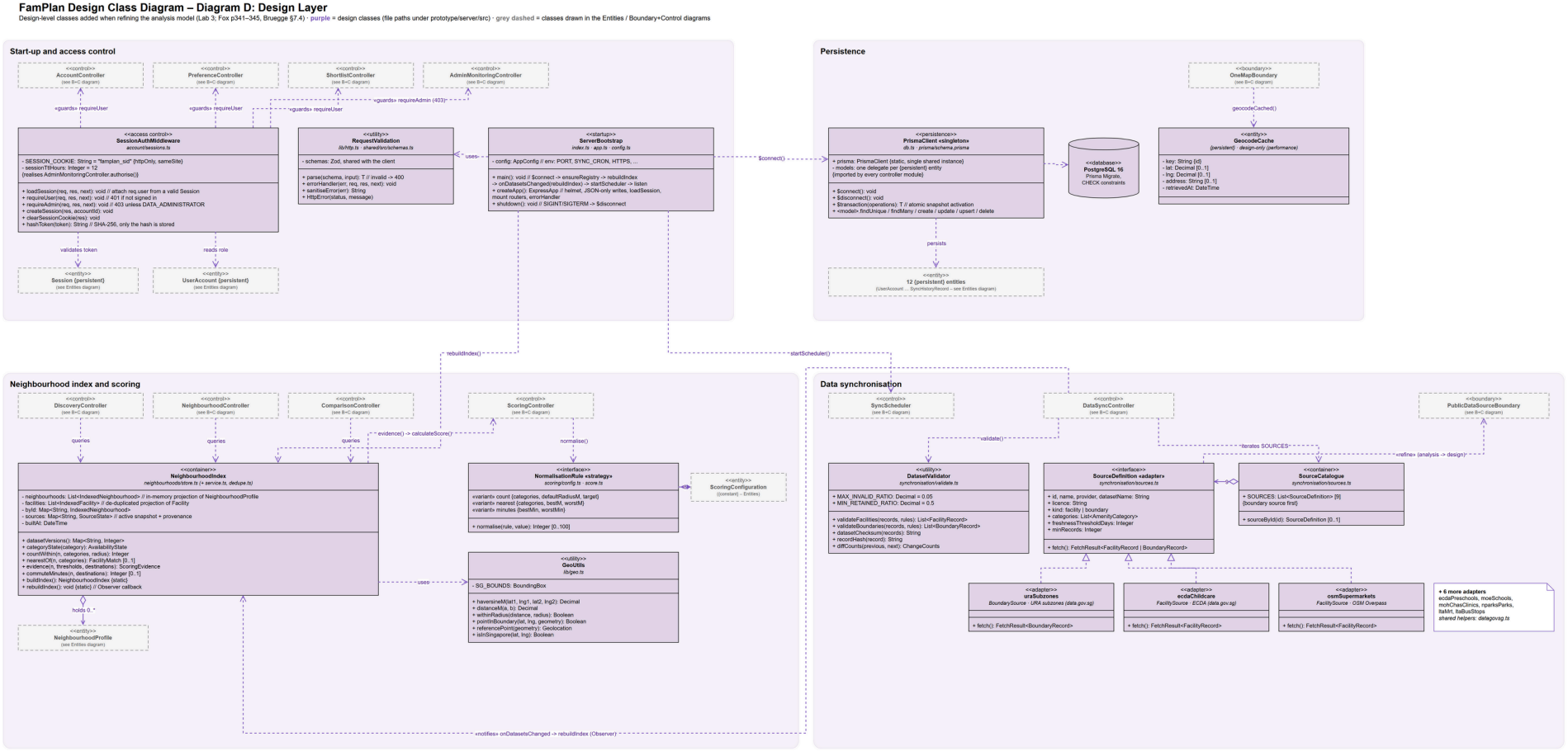
9.3 Robustness View

BCE analysis view illustrating communication between actors, external services, boundaries, controls and entity stores.


FamPlan Class Diagram v1.1 — Diagram B (robustness view): Boundary · Control · Entity
Actor → boundary → control → entity, one band per owner · 10 boundary, 11 control and the entity classes they use (entity details in Diagram A)
ACTORS
BOUNDARY
CONTROL
ENTITY
EXTERNAL SYSTEMS
APOLLO – Account & Authentication
UC-1.1 Register Account · UC-1.2 Login and Logout · UC-1.3 Reset Password · UC-1.4 Manage Account Details · UC-1.5 Delete Account
Visitor
Registered User
AccountBoundary
AccountController
PasswordCredential
PasswordResetToken
Session
UserAccount
EmailDeliveryBoundary
Email Service
«external»
RAPH – Preferences & Discovery
UC-2.1 Manage Family Preferences · UC-2.1 Search Neighbourhoods · UC-2.2 Apply Saved Preferences
Registered User
Visitor
PreferenceBoundary
DiscoveryBoundary
PreferenceController
DiscoveryController
FamilyPreferenceProfile
SearchQuery
AmenitySummary
FamilySuitabilityScore
NeighbourhoodProfile
RouteEstimate
ILHAM – Neighbourhood Map & Route
UC-4.1 View Neighbourhood Profile · UC-4.2 Explore Family Amenities · UC-5.1 View Map and Nearby Places · UC-5.2 Check Commute Route
Visitor
MapRouteBoundary
NeighbourhoodBoundary
RouteController
NeighbourhoodController
to ScoringController (Athena band) ↓
Geolocation
PriorityDestination
RouteEstimate
OneMapBoundary
OneMap
«external»
AmenitySummary
Facility
FacilityMatch
NeighbourhoodProfile
ATHENA – Decision Support & Scoring
UC-6.1 Calculate Family Suitability Score · UC-6.2 Compare Neighbourhoods · UC-6.3 Manage Shortlist and Notes
from NeighbourhoodController (Ilham band) ↑
Visitor
Registered User
DecisionSupportBoundary
ScoringController
ComparisonController
ShortlistController
CriterionScoreBreakdown
CriterionWeight
FamilySuitabilityScore
MissingDataTreatment
ScoringConfiguration
NeighbourhoodComparisonSet
ShortlistEntry
ShortlistHome
UserAccount
EDWARD – Automatic Data Synchronisation & Provenance
UC-7.1 Monitor Automatic Data Synchronisation (SyncScheduler runs on a timer; the administrator only monitors)
Data Administrator
AdminMonitoringBoundary
AdminMonitoringController
SyncScheduler
DataSyncController
UserAccount
DataSourceRegistry
SyncHistoryRecord
DataProvenanceStamp
ValidatedDatasetSnapshot
PublicDataSourceBoundary
Configured Public Data Sources
«external»
Legend ☤ Actor · ▭○ Boundary (orange = external system interface, outside FamPlan) · ▭○ Control · ○ Entity · Lines are communication links: actors talk only to boundaries, boundaries to controls, controls to entities, other controls and external boundaries. Red = link between two owners' bands. An actor or entity may appear in more than one band. Registered User specialises Visitor and Data Administrator specialises Registered User (see the use case diagram).

9.4 Design Layer

Design-level classes added when refining the analysis model (Lab 3): a start-up class, container classes, persistence, access control and design-pattern roles. Grey dashed boxes are classes drawn in Sections 9.1 and 9.2; file paths are relative to prototype/server/src.



9.5 Design-to-Code Traceability

The prototype backend is organised by feature module (`prototype/server/src/<feature>/`) rather than by layer. Inside each module, `routes.ts` is the HTTP boundary, the service and logic files hold the control logic, and the Prisma models in `prisma/schema.prisma` hold the entities. The tables below map each design class in Sections 9.1, 9.2 and 9.4 to the code that implements it; paths are relative to `prototype/` unless shown as `server/src`-relative.

Table 9.5.1 Boundary classes

Design class	Implemented in
AccountBoundary	`client/src/pages/SignIn.tsx, PasswordReset.tsx, Account.tsx`
PreferenceBoundary	`client/src/pages/Preferences.tsx`
DiscoveryBoundary	`client/src/pages/Explore.tsx`
NeighbourhoodBoundary	`client/src/pages/Profile.tsx`
MapRouteBoundary	`client/src/pages/Commute.tsx, components/map.tsx`
DecisionSupportBoundary	`client/src/pages/Compare.tsx, Shortlist.tsx, components/ScoreBreakdown.tsx`
AdminMonitoringBoundary	`client/src/pages/Admin.tsx`
EmailDeliveryBoundary	`server/src/account/mailer.ts` – sendMail()
OneMapBoundary	`server/src/travel/onemap.ts` – searchAddress(), geocodeCached(), onemapRoute()
PublicDataSourceBoundary	`server/src/synchronisation/sources.ts, datagovsg.ts` (SourceDefinition adapters)
(all UI boundaries to server)	`client/src/lib/api.ts` – api<T>(path) over JSON/HTTPS

Table 9.5.2 Control classes

Design class	Module (server/src/)	Key functions / endpoints
AccountController	`account/routes.ts`, `passwords.ts`	POST /auth/register, /auth/login, /auth/logout, /auth/forgot-password, /auth/reset-password; PATCH, DELETE /account; Argon2id hashPassword()
PreferenceController	`preferences/routes.ts`, `presets.ts`	GET, PUT /preferences; POST /preferences/suggest-weights; suggestWeights()
DiscoveryController	`neighbourhoods/service.ts`, `routes.ts`	GET /neighbourhoods – searchNeighbourhoods(), nameMatches()
NeighbourhoodController	`neighbourhoods/service.ts`, `routes.ts`	GET /neighbourhoods/:id, /:id/facilities – neighbourhoodDetail(), amenitySummaries(), facilitiesNear()
RouteController	`travel/route.ts`, `neighbourhoods/routes.ts`	GET /neighbourhoods/:id/route – routeEstimate(); GET /geocode
ComparisonController	`neighbourhoods/routes.ts`	GET /compare (compareQuerySchema: 2–4 ids) – summarise() per item
ScoringController	`scoring/score.ts`, `config.ts`	calculateScore(evidence, weights, datasetVersions), normalise(), roundHalfUp()
ShortlistController	`shortlist/routes.ts`	GET, POST /shortlist; DELETE /shortlist/:id; PUT /shortlist/:id/note
AdminMonitoringController	`synchronisation/routes.ts`	GET /admin/sources, /admin/history, /admin/metrics (behind requireAdmin)
SyncScheduler	`synchronisation/scheduler.ts`	startScheduler() (node-cron, SYNC_CRON), nextRunAt(); api/cron.ts (Vercel Cron in production)
DataSyncController	`synchronisation/pipeline.ts`, `validate.ts`	syncAll(), syncSource(), ensureRegistry(), onDatasetsChanged()

Table 9.5.3 Entity classes and persistence decisions

Design class	Decision	Stored as
UserAccount, PasswordCredential, Session, PasswordResetToken	{persistent}	Prisma models of the same name
FamilyPreferenceProfile, PriorityDestination	{persistent}	Prisma models of the same name
CriterionWeight, AmenityPreference	{persistent, embedded}	FamilyPreferenceProfile.w* columns; amenityThresholds (JSON)
NeighbourhoodProfile	{persistent}	Prisma model **Neighbourhood**
Facility	{persistent}	Prisma model Facility (one set per dataset snapshot)
ShortlistEntry, ShortlistNote	{persistent} / {persistent, embedded}	ShortlistEntry; the note is ShortlistEntry.note (max 500 characters)
DataSourceRegistry, SyncHistoryRecord	{persistent}	Prisma models of the same name
ValidatedDatasetSnapshot	{persistent}	Prisma model **DatasetSnapshot**
GeocodeCache (design only)	{persistent}	Prisma model GeocodeCache (cached OneMap postal lookups)
DataProvenanceStamp	{derived}	Built in NeighbourhoodIndex from DatasetSnapshot and DataSourceRegistry
ScoringConfiguration	{constant, versioned}	`scoring/config.ts` SCORING_CONFIG (version string)
SearchQuery, Geolocation, RouteEstimate, FacilityMatch, AmenitySummary	{transient}	Shared DTOs and Zod schemas in `shared/src`
FamilySuitabilityScore, CriterionScoreBreakdown, MissingDataTreatment, NeighbourhoodComparisonSet	{transient}	Computed per request (calculateScore(), / compare)

Table 9.5.4 Design-layer classes (Section 9.4)

Design class	Role (Lab 3 heuristic)	Implemented in (server/src/)
ServerBootstrap	Start-up class	`index.ts, app.ts, config.ts`
SessionAuthMiddleware	Access control	`account/sessions.ts` – loadSession, requireUser, requireAdmin
PrismaClient «singleton»	Persistent data	`db.ts, prisma/schema.prisma`
NeighbourhoodIndex	Container class	`neighbourhoods/store.ts`
SourceCatalogue	Container class	`synchronisation/sources.ts` – SOURCES
SourceDefinition «adapter» + 9 adapters	Adapter pattern	`synchronisation/sources.ts, datagovsg.ts`
NormalisationRule «strategy»	Strategy pattern	`scoring/config.ts, score.ts`
DatasetValidator	Utility	`synchronisation/validate.ts`
RequestValidation	Utility (input validation, errors)	`lib/http.ts, shared/src/schemas.ts`
GeoUtils	Utility	`lib/geo.ts`

The commercial billing and workspace modules were removed when the project narrowed to personal family planning, so they are not modelled.

10. Dialog Map

